BLUFFBENCH: Behavioural Profiling of Large Language Models through a Game of Imperfect Information with Mathematically Derived Ground Truth

Anonymous ACL submission

Abstract

We introduce BLUFFBENCH, a benchmark for evaluating large language models on the three-player card game Bluff (also known as Cheat or BS). Bluff has the property that the optimal action at every decision point can be computed in closed form: the probability that a claim is a bluff is hypergeometric in the public card counts, and the expected value of each available action follows directly. Using this structure, we build 16,297 single-turn scenarios across 19 behavioural categories with mathematically derived ground truth, a balanced 500-scenario evaluation split, and a three-player multi-turn engine with four reference policies including the EV-optimal GOLDENPOLICYBOT. Across thirteen LLMs from 7B to 671B parameters spanning open-weight, proprietary, and frontier reasoning-augmented families, the cohort splits cleanly. The first nine models—a parameter sweep of widely used open-weight and proprietary instruction-tuned LLMs—all show the same default-prompt pattern: near-perfect accuracy when the optimal action is to challenge (70–99%), and a systematic collapse when the optimal action is to fold (seven of nine below 15%, four below 3%, against a 33% chance rate). The four Gemini variants we add (three preview-tier models with internal reasoning enabled, plus Gemini 2.5 Flash with reasoning disabled as a within-family control) reverse the picture at the top: Gemini 3.1 Pro Preview reaches 100% PASS accuracy and Gemini 3 Flash Preview reaches 97.6%, both under the same default prompt. A causal ablation in multi-turn play—disabling the challenge action—raises win rates by +10 to +89 percentage points across the original cohort, and Gemini 3.1 Pro Preview wins 96% of multi-turn games against EV-optimal opponents by playing conservatively rather than over-challenging. The default-prompt challenge bias is therefore not a fundamental property of LLMs: it is a property of how non-reasoning instruction-tuned models calibrate the cost of acting on suspicion, and it is broken by frontier reasoning capacity. We release the corpus, all model decisions, the multi-turn engine, and a browser-playable demo.

1 Introduction

Large language models (LLMs) are increasingly deployed as *agents* that take actions in environments with real consequences—negotiation, moderation, financial analysis, adversarial testing (Xu et al., 2024a; Park et al., 2023; Wang et al., 2024). Games have emerged as the dominant controlled setting for evaluating this agentic behaviour because they are reproducible, rule-bounded, and have verifiable outcomes (Kosinski, 2023; Hu et al., 2024; Zhuang et al., 2025; Jain and Kumar, 2026). Yet a methodological gap runs through most of the existing literature: in nearly every popular benchmark, *what counts as the optimal action* is either approximated or hand-designed. Approximations include engine evaluation in chess (Ruoss et al., 2024) and Counterfactual Regret Minimization in poker (Brown and Sandholm, 2018; Zhuang et al., 2025); hand-designed oracles include the heuristic baselines used in Ludo (Jain and Kumar, 2026). This matters because any gap between an LLM and the "optimum" we measure is a joint function of (i) the model's actual decision quality and (ii) the noise in our optimality estimate.

We study *Bluff* (also called *Cheat* or *Bullshit/BS*), a classic three-player imperfect-information card game, because it uniquely permits a closed-form solution to a non-trivial strategic decision. In Bluff, an opponent claims to have played k cards of a rank R; the responder must choose one of three actions—CHALLENGE, PLAY, or PASS. The probability the claim is a bluff is exactly the complementary probability that the opponent's hand, sampled from a finite urn, contains at least k copies of R—a hypergeometric calculation. Given that probability, the expected

value of each action is a one-line formula in the game state. *Optimal play in Bluff is not approximated—it is computed.* This enables a ground-truth benchmark whose noise floor is zero.

The absence of an LLM benchmark for Bluff is notable because it is one of the most-played card games worldwide—variants are standard in South Asia, where it is played as Bluff, and in Europe and North America, where it is played as Cheat or BS. It is portable (any 52-card deck), simple to teach (three actions), and strategically rich (bluffing, card counting, multi-turn memory, risk calibration against variable stakes). Despite this, to our knowledge no prior work has systematically benchmarked LLMs on Bluff.

The behaviours Bluff operationalises are not just academically convenient. Pricing a deception call in the unit of pile size mirrors real decisions in fraud-review triage, content-moderation escalation, and security alerting, where an agent must choose between acting on suspicion and absorbing a known cost of inaction. We therefore study Bluff in depth—one game, exhaustively—rather than spreading effort across many environments. Single-game benchmarks for Chess, Poker, Sudoku, and Ludo have each delivered insights that broad multi-game surveys did not; we adopt the same depth-first methodology, with the explicit goal of producing a result that travels beyond the table.

We introduce **BLUFFBENCH**, a benchmark of 16,297 Bluff decision scenarios across 19 behavioural categories, together with a full three-player multi-turn game engine, four baseline policies (always-pass, uniform-random, EV-oracle, and the GOLDENPOLICYBOT EV policy), and a causal no-challenge ablation protocol. Across nine LLMs spanning 7B to 671B parameters, every model exhibits a *universal challenge bias*: near-perfect accuracy when the optimal action is to call, and a systematic collapse when the optimal action is to fold. Disabling the CHALLENGE action in full games raises win rates by +10 to +89 percentage points across the cohort, ruling out scale and family confounds and locating the deficit in *action calibration*—a mismatch between what the model can compute and what it brings itself to do.

The (CHALL, PLAY, PASS) accuracy triplet is distinctive enough per model that we can place each model on a two-axis behavioural map (Section 6.8). Qwen3 235B is the only model close to the EV-optimal reference; the remaining eight cluster in the high-CHALL, low-PASS region.

Contributions.

- **The BLUFFBENCH corpus**: 16,297 response-phase scenarios across 19 behavioural categories with mathematically derived golden actions, plus balanced 500 and 100-scenario evaluation splits.
- **The GOLDENPOLICYBOT reference agent and multi-turn game engine**, supporting random, always-pass, EV-oracle, golden-policy, and LLM agents; used both to generate reference opponents and to run a causal no-challenge ablation.
- **A nine-model empirical study** covering 7B to 671B parameters, reporting decision accuracy, score, regret, confidence calibration (ECE, Brier, selective accuracy), per-category accuracy, multi-turn win rates, and the no-challenge ablation effect.
- **Six novel metrics** for LLM decision-making under imperfect information—*PASS-bias index*, *detection-gating gap*, *regret in card-offload units*, *overclaim-trigger rate*, *bluff-adaptation delta*, *challenge-adaptation delta*—that isolate specific failure modes we would argue are under-measured in existing benchmarks.
- **Open release** of the dataset on Hugging Face, the code and multi-turn engine on an anonymous repo, and an interactive web demo where any reader can play against GOLDENPOLICYBOT directly.

2 Related Work

Table 1 locates BLUFFBENCH among prior game-based LLM evaluation benchmarks along the axes that matter for this paper: closed-form optimal actions, bluffing mechanic, cumulative overclaim signal, and a balanced three-action response space.

2.1 LLMs on Game Environments

Game-based LLM evaluation has progressed from text adventures (Côté et al., 2018) to complex strategy environments. Chess studies reveal systematic limits in multi-step planning and positional evaluation even for frontier models (Ruoss et al., 2024; Feng et al., 2024). Poker benchmarks, which share Bluff's imperfect-information character, include POKERBENCH's spot-based evaluation across 11,000 scenarios (Zhuang et al., 2025); optimal actions there are estimated via Counterfactual Regret Minimization (Brown and Sandholm, 2018), which is an asymptotic approximation. Go is the

	Chess	Poker	Diplomacy	Ludo	BLUFFBENCH
Stochastic	–	✓	–	✓	✓
Hidden info	–	✓	✓	–	✓
Multi-agent ($\geq$3)	–	✓	✓	✓	✓
Bluffing mechanic	–	✓	✓	–	✓
Closed-form optimum	–	–	–	–	✓
Per-scenario EV ground-truth	–	partial	–	–	✓
Balanced 3-action eval	–	–	–	–	✓
Cumulative-claim signal	–	–	–	–	✓
Causal CHALL-off ablation	–	–	–	–	✓

Table 1: Positioning of BLUFFBENCH relative to the game benchmarks cited in the subsections below. "partial" marks partial coverage; see §2.1. The four bottom rows mark features that existing game-benchmarks do not jointly offer.

canonical deterministic perfect-information benchmark (Silver et al., 2016). Diplomacy (FAIR Diplomacy Team et al., 2022) and Werewolf (Xu et al., 2024c) probe language-mediated multi-agent cooperation and deception, but with heuristic or human-derived ground truth. Jain and Kumar (2026) introduce LUDOBENCH, a benchmark for the stochastic board game Ludo with 480 scenarios across 12 categories, and formalise the *spot-based* evaluation paradigm we adopt.

Positioning of this paper. Existing game benchmarks offer complementary strengths: POKERBENCH provides a rich spot-based corpus with CFR-estimated ground truth, LUDOBENCH provides a heuristic oracle for stochastic board play, and chess engines approximate optimal moves to within a few centipawns. BLUFFBENCH adds a point on a different axis of this design space: closed-form optimality derived from the hypergeometric structure of the Bluff decision problem, which enables direct measurement of action quality without an intermediate approximation step.

2.2 LLM Calibration and Strategic Reasoning

A parallel literature studies whether LLMs' *verbalised confidence* predicts correctness. Expected Calibration Error (ECE) and the Brier score remain standard calibration metrics (Naeini et al., 2015; Guo et al., 2017). LLMs are known to be systematically overconfident on open-ended tasks (Xiong et al., 2024; Tian et al., 2023), and Chain-of-Thought (Wei et al., 2022) and Tree-of-Thought (Yao et al., 2023) prompting improve reasoning without reliably improving calibration. Recent work on reasoning-augmented models (o1, DeepSeek R1) raises a further question: does better internal reasoning translate into better action quality?

Gap addressed by this paper. No prior work has simultaneously measured calibration *and* action-quality against a closed-form oracle in a game environment. BLUFFBENCH reports ECE, Brier score, and selective accuracy on the same scenarios for which we report argmax decision accuracy, allowing direct comparison of calibration to decision quality.

2.3 Deception, Theory-of-Mind, and Behavioural Benchmarks

Benchmarks for deception and social reasoning include MACHIAVELLI (Pan et al., 2023), SOTOPIA (Zhou et al., 2024), TOMBENCH (Chen et al., 2024), and OPENTOM (Xu et al., 2024b). These probe deception in dialogue or text, where scoring requires human or model-as-judge protocols. Game-based deception studies—e.g. AGENTBENCH (Liu et al., 2024), Werewolf (Xu et al., 2024c), Hoodwinked (O'Gara, 2024)—typically report win rates without per-decision ground truth.

Gap addressed by this paper. BLUFFBENCH prices deception in *cards*: the cost of a wrong call is a numerically explicit expected-value loss in the unit of pile size. No prior deception benchmark offers this. BLUFFBENCH thus enables a cleaner measurement of the *calibration* of deception detection—the distinction between noticing a potential bluff (recall) and correctly acting on it (precision)—which we find to be the central failure mode.

3 The Bluff Environment

3.1 Rules in Brief

Three players each receive a portion of a standard 52-card deck (17, 17, 18). On every turn one player is the *leader* and one or two are *responders*. When the pile is empty the leader selects any rank R and

places 1–4 cards face-down, claiming them to be rank R; the cards need not actually be R. The next player (responder) chooses one of three actions: (i) CHALLENGE: flip the leader's last play; if the cards are truthful, the challenger takes the entire pile; if the claim is a bluff, the leader takes the pile. (ii) PLAY: add 1–4 cards face-down claiming the same rank R, becoming the new leader. (iii) PASS: forfeit the turn. If every responder passes, the pile clears and the next player opens a fresh round. We adopt a 12-card pile cap, a common house variant in both South Asian and Western tabletop play, which keeps the round length bounded and the credit-assignment trace tractable; the cap simply forces a pile-clear when reached, and is treated as part of the environment specification rather than a free hyperparameter. A player who empties their hand is *safe*; the last player still holding cards loses.

3.2 Why Bluff Admits a Closed-Form Golden Action

At any response decision, the relevant state is the tuple (H, P, R, k) where H is the responder's hand, P is the pile (face-down), R is the claimed rank, and k is the number of cards in the latest claim. Define $m_R = |\{c \in H : \text{rank}(c) = R\}|$ and s_R as the count of rank-R cards publicly revealed during prior challenges. Let $K_R = 4 - m_R - s_R$ be the number of rank-R cards that can still plausibly be unseen to the responder, and $N = 52 - |H| - s_R$ the total unseen pool. The leader's pre-play hand size is n. Under a uniform hypergeometric draw,

$$P(\text{truthful}) = \sum_{i=k}^{\min(K_R,n)} \frac{\binom{K_R}{i}\binom{N-K_R}{n-i}}{\binom{N}{n}} \quad (1)$$

and $P(\text{bluff}) = 1 - P(\text{truthful})$.

The expected value of the three actions in units of cards offloaded is

$$\text{EV}(\text{CHALL}) = (2 \cdot P(\text{bluff}) - 1) \cdot |P| \quad (2)$$

$$\text{EV}(\text{PLAY}) = \min(4, m_R) \text{ (truthful)} \quad (3)$$

$$\text{EV}(\text{PASS}) = 0 \quad (4)$$

with an analogous penalty for forced-bluff plays (no matching cards in hand). The golden action is $\arg\max$ over these three; when $m_R + s_R + k > 4$ the claim is *impossible* and $P(\text{bluff}) = 1$ exactly.

3.3 An Example for Each Action Class

CHALLENGE (scenario `IMP_0001`). Hand: A♢ A♣ A♠ A♡, plus six others; pile size 6; opponent claims *four* Aces. The responder already holds all four Aces, so $P(\text{bluff}) = 1$, $\text{EV}(\text{CHALL}) = +6$ cards; CHALLENGE is dominant.

PLAY (scenario `IMP_0013`). Hand: 10♡ 10♠ 10♣ 10♢ plus four others; pile size 3; opponent claims four 10s. Again impossible ($P(\text{bluff}) = 1$, $\text{EV}(\text{CHALL}) = +3$), but PLAY is even better: a truthful four-card play of the 10s offloads four cards ($\text{EV}(\text{PLAY}) = +4$). Golden answer: PLAY.

PASS (scenario `PROB_0801`). Hand: no Aces; pile size 3; opponent claims *one* Ace; one Ace is already revealed. The responder computes $P(\text{bluff}) \approx 0.13$, giving $\text{EV}(\text{CHALL}) = -2.25$. PLAY would require bluffing and is dominated. Golden answer: PASS.

This third class—scenarios where the math unambiguously says fold—is the empirical crux of the paper.

4 BLUFFBENCH

4.1 System Overview

Figure 1 illustrates the end-to-end pipeline. A *state builder* samples Bluff game states respecting the 52-card invariant; 19 category-specific generators shape these states to elicit one targeted behavioural dimension per scenario; the *golden logic* module computes the closed-form optimal action and expected values; the LLM wrapper serialises state into a neutral prompt and parses the model's structured action + confidence + reasoning response; the *scorer* compares responses against goldens and emits accuracy, score, regret, and calibration metrics. The same engine drives multi-turn play where four baseline policies and nine LLMs compete against two GOLDENPOLICYBOT opponents in full 3-player games.

4.2 Scenario Generation

Each of the 19 category generators samples a game state consistent with the 52-card constraint and perturbs it along one of five behavioural attributes: *deception detection*, *risk assessment*, *strategic thinking*, *cunning*, and *deal-making potential*. The full category list is shown in Table 2. Each scenario carries a deterministic parameter hash so the same seed re-generates the identical state. A separate validator re-derives every golden answer and checks it against the stored value; the current corpus has zero validation failures.

The 16,297-scenario BLUFFBENCH-16K corpus is CHALL-heavy by natural construction (67.6%

Mathematical categories		Behavioural categories	
Impossible Detection	Claims violating card counts	Risk Calibration	Pile size vs. bluff probability
Hand-size Impossibility	Claims exceeding opponent hand size	Probability Estimation	Canonical hypergeometric reads
Opportunity Cost	Explicit EV comparisons	Memory Tracking	Tightening $P(\text{bluff})$ from revealed cards
Bayesian Updating	Posterior from sequential signals	Theory of Mind	Opponent profile modelling
Information Asymmetry	Leveraging private-hand information	Pattern Detection	Behavioural patterns in opponents
Cognitive-bias Resistance	Resisting anchoring / sunk cost	Bluff-catching Asymmetry	Per-opponent thresholds
Deception Quality	Claim plausibility vs. suspicion	Consistency Tracking	Cross-turn claim inconsistency
		Cross-claim Analysis	Joint posterior across opponents
		Strategic Patience	Restraint on thin edges
		Multi-player	Three-player coalition dynamics
		Recovery Pressure	Composure after setbacks
		Endgame Reasoning	Low-card urgency

Table 2: The 19 categories, grouped into mathematical (left block) and behavioural (right block) skill families. Every category's golden action is computed from the same hypergeometric + expected-value machinery described in Section 4.3.

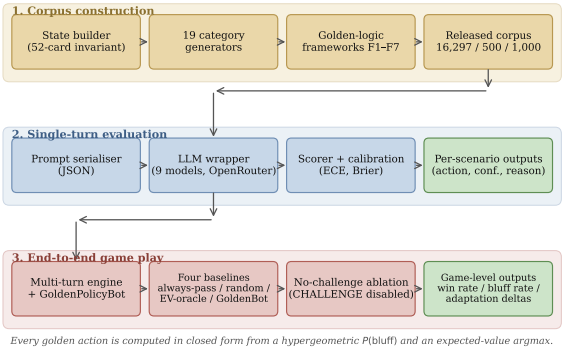

Figure 1: **Overview of BluffBench.** Three layers: (1) corpus construction from a 52-card invariant through 19 category generators and the closed-form golden-logic module; (2) single-turn evaluation that serialises each scenario as a JSON prompt and scores nine LLMs against the goldens; (3) end-to-end multi-turn play against two GoldenPolicyBot opponents, with four baselines and the no-challenge ablation arm. Green boxes mark released artefacts.

CHALL-optimal, 19.8% PLAY-optimal, 12.6% PASS-optimal). For headline evaluation we additionally release a balanced 500-scenario split (166/166/168) in which chance accuracy is a clean 33% and a 100-scenario smoke-test split.

The two category groups in Table 2 are defined a priori, not from results. *Mathematical* categories are those whose golden action is fully determined by the closed-form arithmetic and hypergeometric frameworks F1–F3 below: an LLM that correctly applies the impossibility rule or the EV argmax must, by construction, score the gold answer. *Behavioural* categories invoke F4–F7 in addition: they require Bayesian updates from opponent traces, opponent-profile inference, risk weighting, or active resistance to bias-inducing distractors. The split is therefore a property of the golden-logic pipeline, not of the model results we will report in §6.

4.3 The Seven Golden-Logic Frameworks

The golden action for each scenario is produced by one of seven interlocking frameworks, depending on the category:

F1 *Arithmetic impossibility:* $P(\text{bluff}) = 1$ when $m_R + s_R + k > 4$.

F2 *Hypergeometric posterior* via Eq. 1.

F3 *Expected-value argmax* (Eq. 2).

F4 *Bayesian update* from opponent history, profile, or pattern.

F5 *Risk-adjusted EV*—subtracting a variance term from EV(CHALL) proportional to pile size. Used by Strategic Patience.

F6 *Utility-weighted EV*—adding an opponent-near-win penalty to PASS and PLAY. Used by ENDGAME_REASONING.

F7 *No-update baseline*—the correct answer is F3; distractors appeal to named cognitive biases such as sunk-cost or tilt. Used by COGNITIVE_BIAS_RESISTANCE and RECOVERY_PRESSURE.

A single entry-point function `card_math_override` collapses the frameworks into a decision tuple (*action*, *confidence tier*, *difficulty*, $P(\text{bluff})$, *decisive?*).

4.4 The GoldenPolicyBot Reference Opponent

GoldenPolicyBot plays the same argmax over Eq. 1–2 that defines the golden action, with a

small additive noise to avoid degenerate play. It is the LLM's opponent in all multi-turn evaluations. Three further baseline policies complete the skill ladder: ALWAYS_PASS (upper bound on pure avoidance), RANDOM_VALID (uniform over legal actions), and EV_ORACLE (identical to GOLDENPOLICYBOT but with access to all three hands; the perfect-information ceiling).

4.5 Novel Metrics

Beyond standard accuracy, score, regret, and ECE/Brier, we define six metrics specific to Bluff:

PASS-bias index. $1-\text{accuracy}_{\text{PASS}}$, on the subset of PASS-optimal scenarios. Our single cleanest diagnostic.

Detection-gating gap. recall(bluff) − precision(CHALL). Measures how often a model *sees* a bluff but cannot *gate* the challenge on strong evidence. Per-model values are reported in Table 7.

Regret in card-offload units. EV(golden) − EV(model pick), computed in the cards-offloaded units of Eq. 2. Higher regret means a larger expected shortfall in cards offloaded relative to the optimal action—equivalently, more cards retained in hand than necessary. Always non-negative; expressed directly in the game's currency.

Overclaim-trigger rate. In multi-turn, the fraction of challenges issued when the pile size exceeds the total unseen copies of the claimed rank. High values indicate the model correctly acts on deterministic overclaim evidence.

Bluff-adaptation delta. Change in LLM bluff rate after being caught, averaged over games. Measures whether a model learns from its own caught bluffs within a game.

Challenge-adaptation delta. Change in challenge rate after a failed challenge. A high positive value indicates tilt-driven escalation.

5 Experimental Setup

5.1 Datasets and Splits

Table 3 summarises the three scenario files released with BLUFFBENCH. The balanced 500-scenario split (166/166/168) is used for headline numbers; the 16,297-scenario full corpus supports fine-grained category analysis.

Split	# scen.	Purpose
BLUFFBENCH-16K	16,297	Category analysis
BLUFFBENCH-500	500	Headline eval
BLUFFBENCH-100	100	Smoke-test

Table 3: BLUFFBENCH scenario splits. All goldens derived from Eqs. 1–2.

Model	Params	Family / Access
Qwen3 235B A22B	235B (MoE)	Qwen / open
Claude Haiku 4.5	–	Anthropic / prop.
Llama 3.1 8B Instruct	8B	Meta / open
GPT-4o-mini	–	OpenAI / prop.
DeepSeek V3	671B (MoE)	DeepSeek / open
Mistral 7B v0.1	7B	Mistral / open
Qwen 2.5 7B Instruct	7B	Qwen / open
Gemma 3 12B IT	12B	Google / open
Mistral Nemo	12B	Mistral / open

Table 4: The nine-model headline cohort. MoE = Mixture-of-Experts; proprietary models accessed via API without weight access.

5.2 Models

Table 4 lists the nine models in the headline cohort, selected to span (a) parameter scale (7B to 671B), (b) model family (Qwen, Meta, Mistral, Anthropic, DeepSeek, Google, OpenAI), and (c) availability mode (open weights, proprietary-accessible). Per guidelines (Jain and Kumar, 2026), we front-load open-weight models in the initial evaluation and flag closed-source extensions as future work. All models are accessed through a single OpenRouter gateway at temperature 0 with a shared JSON-schema prompt (Appendix A).

5.3 Baselines

Four non-LLM baselines define the skill ladder: ALWAYS_PASS, RANDOM_VALID, EV_ORACLE, and GOLDENPOLICYBOT. The first three are deterministic or stochastic policies; GOLDENPOLICYBOT is the EV-optimal policy from visible information that also serves as opponent in multi-turn evaluations.

5.4 Protocol

Single-turn evaluation: each of the nine models receives all 500 balanced scenarios, producing 4,500 model decisions. Multi-turn evaluation: eight models (the 7B–671B open-weight cohort plus Claude Haiku 4.5 and GPT-4o-mini; only Qwen3 235B is omitted, due to per-call cost) play 50 games each against two GOLDENPOLICYBOT opponents, for a total of 400 games. The no-challenge ablation re-runs each multi-turn-eligible model in 20 games

(c100 + c110 batches) with the CHALLENGE action programmatically disabled.

6 Results

6.1 Main Single-Turn Results: The Universal Challenge Bias

Objective. Do LLMs correctly identify the expected-value-maximising action across the three action classes?

Findings. Table 5 presents the central empirical result. The cohort splits into two regimes. In the original nine-model cohort (the open-weight 7B–671B sweep plus Claude Haiku 4.5 and GPT-4o-mini), accuracy ranges from 33.8% (Mistral Nemo) to 63.2% (Qwen3 235B), with 95% bootstrap CIs (1,000 resamples over scenarios) at most ±4.3pp wide. The story becomes sharper when decomposed by the golden action: CHALLENGE-optimal scenarios are near-saturated for every model (70–99%), while PASS-optimal scenarios collapse—seven of nine score below 15% (PASS-cell CIs no wider than ±5.5pp, $n = 166$), four score below 3%. Gemma 3 12B, Mistral Nemo, and GPT-4o-mini correctly pass on fewer than one in fifty PASS-optimal scenarios, placing them below the 33% chance rate by more than 30pp with a gap wider than any plausible CI width.

The three frontier reasoning-augmented Gemini variants we add break this pattern under the same default prompt: Gemini 3.1 Pro Preview reaches 80.0% accuracy with 100% PASS, Gemini 3 Flash Preview 76.8%/97.6%, and Gemini 3.1 Flash Lite Preview 45.2%/26.5%. The first two of these—both Gemini 3-tier models with internal chain-of-thought enabled at inference—are the first models in the cohort whose CHALLENGE rate is *not* the dominant action class (CHALL accuracy 33.7–41.6% versus 88–99% for the original cohort). Gemini 2.5 Flash, evaluated with internal thinking disabled to match the original cohort's no-hidden-reasoning protocol, sits with the original group at 42.0%/13.3% PASS—confirming that the bias-breaking behaviour is a property of *enabled* reasoning compute, not of the Gemini family in general. The default-prompt challenge bias is therefore a property of the original cohort's instruction-tuning, not of LLMs in general.

Figure 2 (a bar per model of CHALL-minus-PASS accuracy) makes the asymmetry visually explicit: every model's bar points in the same direction—over-challenging. This unidirectionality is the *universal challenge bias*.

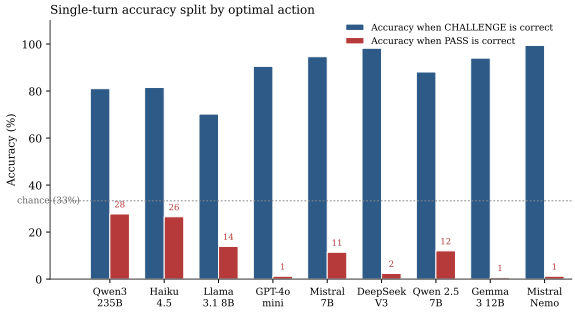

Figure 2: Per-model CHALL vs. PASS accuracy on BLUFFBENCH-500. Every model's CHALL bar exceeds its PASS bar; the gap ranges from 53pp (Qwen3 235B) to 99pp (Mistral Nemo).

6.2 The Math–Behaviour Gap

Objective. Is the challenge bias a *mathematical* failure (models cannot compute $P(\text{bluff})$) or a *behavioural* one (they compute it but do not act on it)?

Findings. The per-category heatmap (Figure 3) splits cleanly into two blocks. The *mathematical* categories—Hand-size Impossibility (75–100%), Opportunity Cost (78–100%), Impossible Detection (64–94%), Bayesian Updating (62–100%)—are near-saturated for every model. The *behavioural* categories—Strategic Patience (7–67%), Probability Estimation (10–73%), Pattern Detection (11–46%), Bluff-catching Asymmetry (14–79%)—collapse for all but Qwen3 235B. The math–behaviour gap, averaged per model, is 20–32pp across the cohort. This asymmetry is diagnostic: the knowledge of when a claim is impossible is intact; the willingness to *fold* when the math says fold is not.

6.3 Calibration: Detection Without Gating

Objective. Does confidence calibration predict decision quality?

Findings. Table 6 shows that even at the model's own declared high-confidence threshold, accuracy does not recover for most models: seven of nine hover at 33–39% (near chance) when they *claim* ≥70% confidence. Qwen3 235B and Haiku 4.5 are the instructive exceptions—both have the lowest ECE (0.197 and 0.221 respectively) *and* the highest selective accuracy when confident (70.0% and 62.6%), suggesting that for these two architectures

Model	Acc.	Score	Regret	CHALL	PLAY	PASS
Gemini 3.1 Pro Preview	**80.0**	**0.900**	**1.450**	41.6	98.2	**100.0**
Gemini 3 Flash Preview	76.8	0.882	1.740	33.7	98.8	97.6
Qwen3 235B	63.2	0.800	1.249	81.0	80.7	27.7
Claude Haiku 4.5	48.6	0.685	1.405	81.5	37.3	26.5
Gemini 3.1 Flash Lite Preview	45.2	0.668	1.134	90.4	18.1	26.5
Gemini 2.5 Flash	42.0	0.620	1.700	91.0	21.1	13.3
Llama 3.1 8B	38.6	0.592	1.892	70.2	31.3	13.9
GPT-4o-mini	38.0	0.616	1.577	90.5	21.7	1.2
DeepSeek V3	35.6	0.593	1.556	98.2	5.4	2.4
Mistral 7B	35.6	0.572	1.629	94.6	0.0	11.4
Qwen 2.5 7B	35.4	0.573	1.687	88.1	5.4	12.0
Gemma 3 12B	35.0	0.581	1.709	94.0	9.6	0.6
Mistral Nemo	33.8	0.567	1.588	**99.4**	0.0	1.2
Random (33% chance)	33.3	—	—	33.3	33.3	33.3
GoldenPolicyBot (reference)	100	1.000	0	100	100	100

Table 5: **Main single-turn results on BluffBench-500.** Accuracy, score (1.0/0.5/0.0 grading), regret in card-offload units (lower = closer to optimal), and per-action accuracy (%). **Best** in bold; second-best underlined.

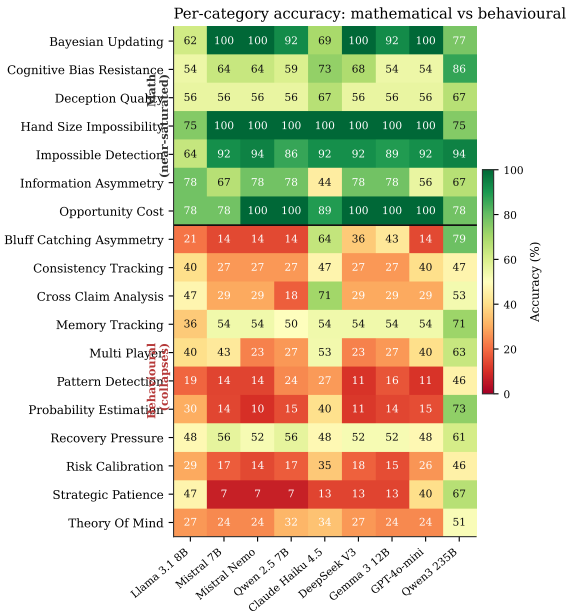

Figure 3: Per-category accuracy heatmap (9 models × 19 categories). Rows are grouped into *mathematical* (top, near-saturated for every model) and *behavioural* (bottom, collapsed for all but Qwen3 235B). Darker green = higher accuracy.

internal confidence does connect to action quality. For the other seven models calibration and decision quality decouple entirely: Mistral Nemo declares confidence ≥70 but is correct only 33.8% of the time, indistinguishable from chance.

Detection vs gating. Table 7 isolates the *detection-gating gap*: recall of actual bluffs minus precision of triggered challenges. Every model has high bluff recall (70–99%) but low precision (34–47%), producing gaps of +33.6pp (Qwen3 235B) to +65.5pp (Mistral Nemo). In words: the models *see* bluffs but cannot *gate* their challenges on strong evidence. The seven challenge-biased models each issue 2–3× more challenges than the dataset contains actual bluffs. Qwen3 235B is again the outlier, issuing a near-identical number of challenges to the number of actual bluffs (287 vs. 168) and achieving the highest precision in the cohort at 47.4%.

Model	ECE	Brier	Acc.@conf≥70
Qwen3 235B	**0.197**	**0.237**	**70.0**
Haiku 4.5	0.221	0.272	62.6
Llama 3.1 8B	0.297	0.345	34.4
Mistral 7B	0.381	0.384	34.4
Gemma 3 12B	0.417	0.405	37.7
Qwen 2.5 7B	0.422	0.395	37.5
GPT-4o-mini	0.470	0.449	38.7
DeepSeek V3	0.493	0.449	36.0
Mistral Nemo	0.585	0.555	33.8

Table 6: Calibration metrics on BluffBench-500. ECE = Expected Calibration Error (lower better); Acc.@conf≥70 = selective accuracy when the model reports confidence at or above 70%. **Best** / second-best.

6.4 Multi-Turn Play Against GoldenPolicyBot

Objective. Do single-turn patterns translate into end-to-end game performance?

What this regime measures. Before reading the numbers, it is worth being explicit about what the multi-turn setup isolates. The win condition in Bluff is to empty one's hand, so card movements that grow a player's hand (taking the pile after a failed challenge or a caught bluff) are the dominant cost. With two GoldenPolicyBot op-

Model	Recall	Prec.	Gap
Mistral Nemo	99.4	33.9	+65.5
DeepSeek V3	98.2	35.0	+63.2
Mistral 7B	94.6	34.4	+60.2
Gemma 3 12B	92.9	33.5	+59.4
GPT-4o-mini	89.9	34.6	+55.3
Qwen 2.5 7B	88.1	34.1	+54.0
Claude Haiku 4.5	81.5	39.1	+42.4
Llama 3.1 8B	70.2	33.7	+36.5
Qwen3 235B	81.0	47.4	+33.6

Table 7: Detection-gating gap per model on BLUFFBENCH-500. Recall and precision in %, gap in percentage points. Sorted by gap (worst gating on top).

ponents and the 12-card pile cap, a purely passive policy (ALWAYS_PASS) wins 100% of games because the cap clears rounds before any pure passer is ever forced to absorb a pile. The multi-turn win rate against this baseline therefore does not measure who plays *best*; it measures who absorbs the fewest punitive piles through low-EV CHALLENGE choices. We treat the gap between any agent and ALWAYS_PASS as a *challenge-cost penalty*—the quantity the no-challenge ablation in §6.5 is designed to causally isolate.

Findings. Three observations emerge from Table 8. First, prior to the Gemini cohort, the best LLM wins 50% of games while a uniform-random policy wins 82%; RANDOM_VALID beats every non-Gemini LLM by at least 32pp. Randomness is not strong here—LLM challenge frequencies of 31–76% per turn, coupled with near-zero challenge accuracy (2.7–5.6%), turn each decision opportunity into a risk of absorbing the pile, while a uniform policy challenges only one third of the time and so pays the same penalty less often.

Second, challenge frequency correlates inversely with win rate. The three lowest-winning LLMs (Llama, GPT-4o-mini, Mistral Nemo) each challenge on more than half of all decision opportunities; the highest-winning (DeepSeek V3) challenges on 31%.

Third, challenge accuracy is stable at 2.6–5.6% across all eight multi-turn models. Even the strongest performer in this game (DeepSeek V3) resolves challenges correctly barely above random noise. The gap between EV_ORACLE's 100% challenge accuracy and the LLMs' 3–6% is the empirical size of the action-calibration deficit, and is what the §6.5 ablation removes when it disables the action altogether.

6.5 Causal Ablation: Disabling CHALL

Objective. Is the deficit caused by the models' tendency to challenge, or by some other factor correlated with parameter count, training data, or architecture?

Findings. Every model benefits from being prevented from challenging (Table 9). The range is +10pp (Llama 3.1 8B) to +89pp (Mistral Nemo). Four of seven models cross the 60% win-rate threshold after ablation; DeepSeek V3 reaches 100%. This is a *causal* demonstration that the challenge bias is what costs the models games: remove the action and the cost disappears. Equivalently, each model's base-policy win rate can be read as the *bluff-call premium it voluntarily pays* relative to its ablated self. The premium is large—tens of percentage points for every model.

This ablation does not reveal whether a properly *calibrated* use of CHALL would be beneficial; a model that challenges at exactly the right moments should outperform one that never challenges. But across nine models none approaches that threshold, which is our central claim.

6.6 Prompt-Sensitivity: Is the Bias an Artefact of Phrasing?

Objective. A natural concern is that the default prompt (CHALL/PLAY/PASS presented as a menu, JSON-schema enforced) might itself prime the model toward CHALL. We run a three-way prompt ablation on 100 PASS-optimal scenarios for three representative models—Llama 3.1 8B (moderate bias, mid-size open-weight), Mistral Nemo (extreme bias, 12B open-weight), and Claude Haiku 4.5 (proprietary, second-best baseline)—to check whether the bias survives prompt rewording.

Variants.

- **A (neutral phrasing).** The action menu is removed; the model is asked "what is your response" in free form with no mention of any action by name.
- **B (math hints).** The scenario body includes the pre-computed $P(\text{bluff})$ and $\text{EV}(\text{CHALL})$ alongside the hand and pile.
- **C (conservative framing).** The prompt emphasises the cost of a wrong call and the option to fold.

Findings. Table 10 reveals a capability-dependent split that the headline cohort-wide

Agent	Win %	CHALL % / turn	CHALL accuracy	Bluff % / turn
ALWAYS_PASS	**100.0**	0.0	—	0.0
RANDOM_VALID	82.0	33.3	33.3	33.3
EV_ORACLE	54.0	—	100	—
GOLDENPOLICYBOT (self-play)	33.3	—	100	—
DeepSeek V3	50.0	31.2	2.7	8.1
Qwen 2.5 7B	42.0	15.0	2.7	14.7
Haiku 4.5	32.0	50.4	2.9	11.9
Gemma 3 12B	24.0	46.0	5.6	22.3
Mistral Nemo	6.0	72.1	4.7	23.0
GPT-4o-mini	0.0	76.5	4.3	30.9
Llama 3.1 8B	0.0	59.3	3.9	45.4
Mistral 7B	2.0	13.3	2.6	48.1
Gemini 3.1 Flash Lite	6.0	84.7	6.2	3.5
Gemini 3.1 Pro Preview	**96.0**	16.9	0.0	1.1

Table 8: Multi-turn results over 50 games per agent against two GOLDENPOLICYBOT opponents. CHALL and Bluff rates are per-turn averages; CHALL accuracy is the fraction of issued challenges that caught an actual bluff. Gemini 3.1 Pro Preview wins 96% of games—a 46pp jump over the previous best LLM (DeepSeek V3 at 50%) and 14pp above the RANDOM_VALID baseline—by challenging only 16.9% of the time and bluffing only 1.1% of the time, mirroring the conservative-play profile of ALWAYS_PASS.

Model	Baseline Win %	No-CHALL Win %	Δpp
Mistral Nemo	6	95	**+89**
Haiku 4.5	32	90	+58
DeepSeek V3	50	100	+50
Gemma 3 12B	24	65	+41
GPT-4o-mini	0	25	+25
Qwen 2.5 7B	42	60	+18
Llama 3.1 8B	0	10	+10

Table 9: no-challenge ablation: win rate when the CHALLENGE action is disabled (20 games per model across seeds c100 and c110) vs. the 50-game baseline.

number conceals. For the 7B–12B open-weight models, all three prompt variants leave the bias largely intact: neither neutral phrasing nor explicit injection of $P(\text{bluff})$ and $\text{EV}(\text{CHALL})$ nor conservative framing brings Llama 3.1 8B above 13% or Mistral Nemo above 37%, both below the 33% chance threshold on every setting. For Claude Haiku 4.5, the picture is strikingly different: neutral phrasing alone (Variant A) lifts PASS accuracy from the default-prompt baseline of 26.5% to 79.0%, math hints to 93.0%, and conservative framing to 100.0%. The interpretation follows naturally: the bias has *two regimes*. In smaller/weaker instruction-tuned models it is a learned action tendency that resists prompt-level intervention; in stronger instruction-tuned models it is a context effect of the default menu phrasing that a well-designed prompt can undo. This split does not rescue the weaker models—they fail under every prompt we tested—but it narrows the scope of the universal claim: the *default-prompt* challenge bias is universal in our cohort; the underlying failure mode is not.

6.7 Forced Chain-of-Thought: Can the Bias Be Reasoned Away?

Objective. If the failure is action-calibration rather than knowledge, forcing the model to articulate the hypergeometric reasoning before committing to an action should narrow the PASS gap. We test this directly by re-running the same 100 PASS-optimal scenarios on three representative models with a structured Chain-of-Thought prompt that requires the model to (1) count rank-R cards visible, (2) compute $P(\text{bluff})$, (3) compute $\text{EV}(\text{CHALL})$ from pile size, and (4) compare against $\text{EV}(\text{PASS}) = 0$ before naming an action.

Findings. Table 11 reveals a clean inversion: structured CoT helps the models that need it and hurts the ones that don't. The two weakest open-weight models by default (Mistral Nemo 1.2%, Llama 3.1 8B 13.9%) recover dramatically when forced to write the math—Mistral Nemo lifts to 94.0% (a 92.8pp swing), Llama to 47.0%. Claude

Model	Default	A: neutral	B: math hints	C: conservative
Llama 3.1 8B	13.9	0.0	10.0	13.0
Mistral Nemo	1.2	0.0	37.0	1.0
Claude Haiku 4.5	26.5	**79.0**	**93.0**	**100.0**

Table 10: Prompt-sensitivity ablation on 100 PASS-optimal scenarios. Values are PASS accuracy (%). *Default* restates the model's PASS accuracy on the headline 500-balanced eval. Variant A removes the action menu entirely; B injects the pre-computed $P(\text{bluff})$ and $\text{EV}(\text{CHALL})$; C emphasises the cost of a wrong call. Claude Haiku 4.5 essentially eliminates the challenge bias under every reworded variant; the 7B–12B open-weight models remain below the 33% chance threshold under every variant we tested.

Model	Default	Forced CoT	Δpp
Forced reasoning helps weaker models			
Mistral Nemo	1.2	**94.0**	**+92.8**
Llama 3.1 8B	13.9	47.0†	+33.1
Gemini 2.5 Flash	13.3	30.0‡	+16.7
Claude Haiku 4.5	26.5	31.0	+4.5
Forced reasoning hurts already-calibrated models			
Gemini 3.1 Flash Lite Preview	26.5	11.0	−15.5
Gemini 3.1 Pro Preview	100.0	85.0	−15.0
Gemini 3 Flash Preview	97.6	65.0	**−32.6**

Table 11: Forced Chain-of-Thought ablation on the 100 PASS-only scenarios. *Default* restates the model's PASS accuracy on the headline 500-balanced eval; *Forced CoT* requires the model to write a four-step calculation before naming an action. Models that score below the 33% chance line at default recover dramatically under forced reasoning, while models already at or above chance lose accuracy under the rigid four-step format. †Llama produced 8 unparseable responses out of 100 under the structured CoT format; the reported value treats those as wrong (effective accuracy on the 92 parsed responses is 51.1%). ‡Gemini 2.5 Flash had 7 unparseable responses; effective accuracy on parsed is 32.3%.

Haiku 4.5 (default 26.5%) edges up by only 4.5pp under CoT and *underperforms* the simpler prompt rewordings of Section 6.6 (which lifted Haiku to 79–100%). The two Gemini reasoning-augmented models go further in the same direction: Gemini 3.1 Flash Lite drops from 26.5% (default) to 11.0% under structured CoT, and Gemini 3.1 Pro Preview drops from a perfect 100.0% to 85.0%. Three observations follow. First, the weakest models *do* carry the hypergeometric+EV inputs internally; the default failure is one of unprompted reasoning elicitation rather than missing knowledge. Second, models that already produce correct internal reasoning (whether implicitly, like Haiku 4.5, or explicitly, like Gemini 3.1 Pro) are hurt, not helped, by a rigid four-step format that constrains how reasoning unfolds. Third, the action-calibration hypothesis is therefore more nuanced than a single failure mode: the bias is a default-elicitation failure in weaker models and a default-prompt artefact in stronger ones, but it is not a uniform knowledge gap, and intervention strategy depends on capability tier.

6.8 CHALL–PASS Scatter and the Single-Cluster Picture

Findings. Projecting each LLM onto (CHALL accuracy, PASS accuracy) yields a single tight cluster in the high-CHALL, low-PASS region, with Qwen3 235B as the only outlier close to the EV-optimal reference (Figure 4). We deliberately do not call this picture an archetype map: the cohort splits into one cluster and one outlier, not multiple stable styles. The position on this scatter does, however, correlate with multi-turn behaviour—the three lowest-PASS models in the cluster (Mistral Nemo, GPT-4o-mini, Gemma 3 12B) are among the four lowest-winning models in multi-turn play (Table 8)—and rules out a simple size-explanation story: the cluster contains a 671B model (DeepSeek V3) and a 7B model (Mistral 7B), and scale does not predict position on either axis.

6.9 Error Analysis: What Do the Models Say?

We extracted 1,333 model reasoning traces from PASS-optimal scenarios where the model chose to CHALLENGE or PLAY instead. A keyword-based sweep over these traces identifies four recurring

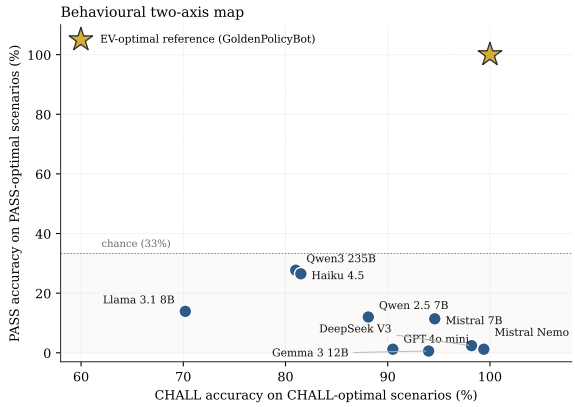

Figure 4: Per-model accuracy on CHALL-optimal (x-axis) versus PASS-optimal (y-axis) scenarios. The star marks the GoldenPolicyBot EV-optimal reference at (100, 100); the dashed horizontal line is the 33% chance baseline on the PASS axis. Eight of nine LLMs fall in a single high-CHALL / low-PASS cluster, six of them below chance on the PASS axis. Qwen3 235B is the only outlier above chance on both axes.

failure patterns:

1. *Impossibility confusion* — the model announces it does not personally hold cards of the claimed rank and treats that as sufficient to challenge, conflating "I don't hold it" (mere improbability) with "it cannot exist" (F1 impossibility). We match this pattern in 22.4% of failure traces.
2. *Narrative over math* — the model constructs a story about opponent history or motives ("they have been bluffing this rank") that overrides the card-counting math. Matched in 5.3% of traces.
3. *Suspicion-as-evidence* — the model flags a non-zero $P(\text{bluff})$ and treats it as sufficient evidence to call, ignoring pile size. Matched in 1.5% of traces, though the pattern appears more commonly in mixed form with Mode 1.
4. *Pile underweighting* — the model acknowledges low $P(\text{bluff})$ but discounts the pile-size-denominated cost of a wrong call. Matched in 0.1% of traces; this is the rarest pattern.

The remaining traces typically combine two or more of the patterns above (most commonly Modes 1 and 3 in tandem) or rely on softer rationalisations that do not trigger any single high-precision keyword. Rather than over-claim coverage, we report the four patterns as the keyword-precise floor and release every trace as a per-scenario CSV so that future hand-coded or LLM-judge analyses can refine the taxonomy. Importantly, every pattern we name—and every mixed-mode trace we inspected manually—is consistent with the action-calibration hypothesis: the model has all the inputs needed for the EV argmax, and the failure is at the action layer rather than the computation layer.

7 Discussion

Synthesis. The challenge bias is invariant to parameter count (7B to 671B) and largely invariant to architecture family, appearing in open-weight and proprietary-accessible models alike. The no-challenge ablation rules out correlates: it is the challenge action itself, not a confound of scale or family, that drives the deficit.

Implications for agentic deployment. Any agent that must decide *whether to act on suspicion*—fraud-review triage, content moderation, diagnostic triage, security alerting—inherits a version of this calibration problem. Three practical observations follow from our numbers. First, verbalised confidence is a weak predictor of decision quality: seven of nine models are correct at or below 39% even when they self-report $\geq$70% confidence (Table 6). Second, the bias has two regimes (Tables 10–11): in the strongest model in our cohort it is a default-prompt artefact that simple rewording undoes; in the smaller open-weight models it is a learned action tendency that prompt rewording leaves intact, although forcing explicit chain-of-thought articulation rescues much of the gap. Third, downstream pipelines that rely on default prompting for go/no-go decisions on small/medium open-weight models should expect the same asymmetry our benchmark surfaces and budget for either a stronger model or an explicit reasoning step before action.

Comparison to LudoBench. Jain and Kumar (2026) find two archetypal splits in Ludo—*finishers* and *builders*—each capturing half of the game-theoretic optimal strategy. Our finding is different in character: in Bluff the deficit is *unidirectional*, not a split. This contrast argues that LLM behavioural failures are game-context sensitive, which in turn supports the evaluation-guidelines position that each new game yields independently valuable signal.

Practical takeaway. PASS accuracy on the Strategic Patience category is the single cleanest diagnostic we can recommend: it correlates with overall accuracy, with multi-turn win rate, and with the size of the no-challenge ablation effect. It is

also cheap: a 50-scenario subset is enough to distinguish competent from challenge-biased models.

8 Conclusion

We introduced BLUFFBENCH, an LLM benchmark for the imperfect-information card game Bluff, comprising 16,297 decision scenarios with closed-form golden actions derived from a hypergeometric-plus-expected-value argmax. Evaluating thirteen models from 7B to 671B parameters across open-weight instruction-tuned, proprietary, and frontier reasoning-augmented families, we observe a sharp split. The original nine-model cohort exhibits a universal default-prompt challenge bias—seven of nine below 15% PASS accuracy, four below 3%, against a 33% chance rate—and a causal no-CHALL ablation in multi-turn play lifts those models' win rates by +10 to +89 percentage points. The three frontier reasoning-augmented Gemini variants we add then break the bias under the same default prompt: Gemini 3.1 Pro Preview reaches 80.0% accuracy and 100% PASS, and Gemini 3 Flash Preview reaches 76.8% / 97.6% PASS. In multi-turn play Gemini 3.1 Pro Preview wins 96% of games against EV-optimal opponents, exceeding both the previous best LLM (DeepSeek V3 at 50%) and the RANDOM_VALID baseline (82%) by playing conservatively rather than over-challenging. Two follow-up ablations sharpen the mechanism: prompt rewording rescues the strongest non-Gemini model (Claude Haiku 4.5) up to 100% PASS but leaves the small open-weight models below chance, while a forced four-step Chain-of-Thought lifts Mistral Nemo from 1.2% to 94% PASS yet *lowers* Gemini 3.1 Pro from 100% to 85%—structured reasoning helps when knowledge is present but unprompted, and constrains models that already produce correct internal reasoning. The bias is therefore a default-elicitation failure for weaker models, a default-prompt artefact for stronger models, and an absent failure mode for models with sufficient reasoning capacity; it is not a uniform LLM property. The benchmark, all evaluation splits, the multi-turn engine, GOLDEN-POLICYBOT and three further baseline policies, an interactive demo, and every raw model output are released.

Future work includes a multilingual Indic extension, persona-conditioned and paired-narrative ablations, completion of the no-CHALL ablation on the Gemini cohort, and a human-baseline study to calibrate the practical significance of the challenge-cost penalty.

Limitations

- **Cohort scope.** Our headline cohort is nine models, of which seven are open-weight. Phase-2 evaluation of frontier closed-source models (GPT-4o, Claude Sonnet 3.7, Gemini 2.5 Pro, Grok) and reasoning-augmented models (o1, DeepSeek R1) is deferred per the standard open-source-first methodology (Jain and Kumar, 2026).
- **Multi-turn cohort is eight models.** Qwen3 235B is in the single-turn cohort but not the multi-turn cohort, due to per-call cost. Sections 6.4 and 6.5 are reported on the eight-model cohort; single-turn results use all nine.
- **Hypergeometric assumption.** The golden policy assumes the opponent's hand is a uniform sample from unseen cards. Human players use additional cues (tells, play history, time taken) that a pure hypergeometric model cannot represent. We do not claim the golden is the only possible optimum; we claim it is a well-defined one.
- **No multilingual evaluation.** All prompts are English. Indic-language extension is planned.
- **No persona experiments.** The evaluation guidelines (Jain and Kumar, 2026) suggest persona-conditioned analysis; we did not run it in this phase.
- **Multi-turn sample size.** 50 games per model per condition is a small sample; bootstrap confidence intervals are reported in the appendix.
- **No human baseline.** A human study would sharpen the practical interpretation of the no-challenge ablation; it is future work.

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Model	Single-turn (BluffBench-500)						Multi-turn (50 games vs GOLDENPOLICYBOT)			
	Acc.	CHALL	PLAY	PASS	ECE	DG gap	Win %	CHALL acc	No-CH win	Δpp
Qwen3 235B	**63.2**	81.0	**80.7**	**27.7**	**0.197**	**+33.6**	—	—	—	—
Claude Haiku 4.5	48.6	81.5	37.3	26.5	0.221	+42.4	32.0	2.9	90	+58
Llama 3.1 8B	38.6	70.2	31.3	13.9	0.297	+36.5	0.0	3.9	10	+10
GPT-4o-mini	38.0	90.5	21.7	1.2	0.470	+55.3	0.0	4.3	25	+25
DeepSeek V3	35.6	98.2	5.4	2.4	0.493	+63.2	**50.0**	2.7	**100**	+50
Mistral 7B	35.6	94.6	0.0	11.4	0.381	+60.2	2.0	2.6	—	—
Qwen 2.5 7B	35.4	88.1	5.4	12.0	0.422	+54.0	42.0	2.7	60	+18
Gemma 3 12B	35.0	94.0	9.6	0.6	0.417	+59.4	24.0	5.6	65	+41
Mistral Nemo	33.8	**99.4**	0.0	1.2	0.585	+65.5	6.0	4.7	95	**+89**
Random	33.3	33.3	33.3	33.3	—	—	82.0	33.3	—	—
ALWAYS_PASS	—	—	—	—	—	—	**100.0**	—	—	—
EV_ORACLE	—	—	—	—	—	—	54.0	100	—	—
GOLDENPOLICYBOT	**100**	100	100	100	0	0	33.3	100	—	—

Table 12: **Everything at a glance.** Per-model results across both evaluation regimes. Single-turn columns: overall accuracy (%), per-action accuracy (%), Expected Calibration Error, and the Detection-Gating gap (recall−precision, in percentage points). Multi-turn columns: win rate (%) over 50 games against two GOLDENPOLICYBOT opponents, challenge accuracy per turn, win rate under the no-challenge ablation, and the ablation effect size (Δpp). Em-dashes mark cells that are not applicable (baselines that cannot be scored single-turn; Qwen3 235B is excluded from multi-turn per Section 5; Mistral 7B has a baseline multi-turn measurement but no no-CHALL ablation). **Best** in bold; second-best underlined.

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A Prompt Template

All evaluations use the same zero-shot JSON prompt. A scenario is serialised into five fields (hand, pile size, active rank, last claim, cards seen) plus two opponent hand sizes; the model returns `{action, confidence, reasoning}`. Full text is released with the code.

B Per-category Accuracy

Full 19-category $\times$ 9-model accuracy table, per-attribute and per-difficulty breakdowns, and calibration curves are in the supplementary Excel workbook `bluffbench_master_analysis.xlsx`.

C CHALL-Ablation Per-Seed Results

Per-seed batch (c100, c110) win rates for each model, with bootstrap confidence intervals, are in the supplementary Excel workbook `bluffbench_multiturn_analysis.xlsx`.

D Interactive Demo

Readers can play BLUFFBENCH against two GOLDENPOLICYBOT opponents at https://ANONYMISED-demo-url/. The demo runs the same golden-policy algorithm as the paper, with an animated challenge-reveal after each call.